

The University of Azad Jammu and Kashmir, Muzaffarabad.



Computer Networks

Department of Software Engineering

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Task Interpretation

The uploaded open-ended lab manual asks for the design of a reliable and scalable network for a newly established educational institute serving administration and academic departments. The required deliverables are requirements analysis, topology justification, cabling plan, switch configuration, router configuration, IP addressing table, network diagram, and future recommendations.

1. Network Requirements Analysis

1.1 Scenario

A newly established educational institute requires a small computer network connecting its administration and academic departments. The network should provide reliable communication, logical separation of departments, straightforward management, and sufficient capacity for future expansion.

1.2 Estimated Devices

Department / Area	PCs	Other Devices	Total Estimated
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Administration	8	1 network printer	9
Faculty / Academic	12	1 network printer	13
IT / Lab	20	1 server	21
Network Infrastructure	—	3 switches + 1 router	4

The design therefore provides connectivity for approximately 47 endpoint/infrastructure devices. The exact device count is an assumed design value for this open-ended scenario and should be adjusted if your instructor expects a different institute size.

TASK 1 — Network Requirements Verification

For the packet tracer implementation, the network is designed as:

Switches corresponding to the departments.

1. Administration
2. Academic
3. IT/Lab

Each with 3 PCs initially.

So one router, 3 switches and 9 PCs (3 for each switch) result in a total of **13 devices**.

This allows departments to communicate through the router while keeping their networks logically separated. It's a scaled implementation of the proposed institute network.

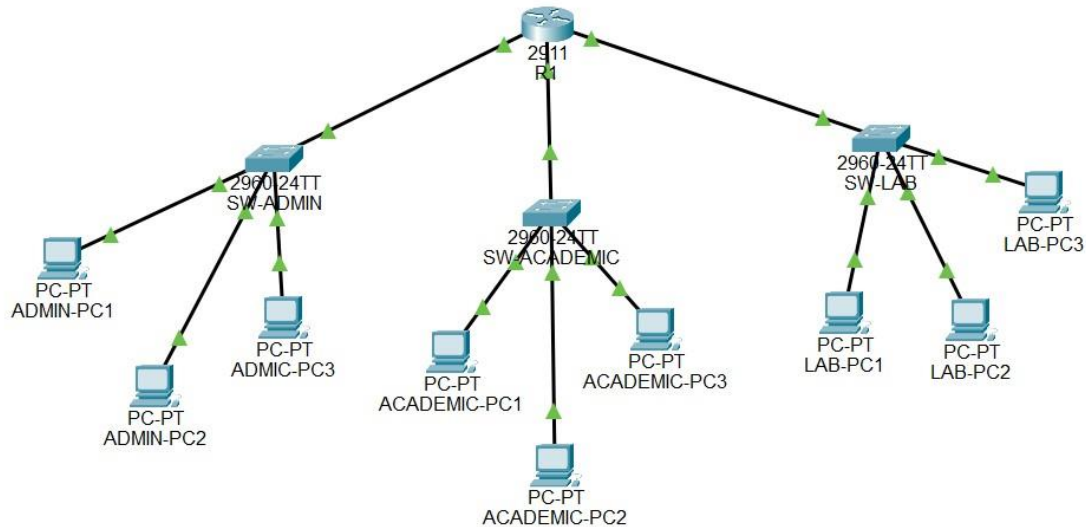
2. Topology Selection and Justification

A hierarchical star topology is selected. Each department connects its end devices to an access switch, and the departmental switches connect toward a central Layer-3 routing point. This arrangement is suitable for a small institute because individual links can be isolated for troubleshooting, expansion can be made by adding ports or switches, and the logical departmental networks can be separated using different IP subnets.

The design uses three access switches: one for Administration, one for Faculty/Academic users, and one for the IT/Lab area. A router provides inter-network routing between the departmental subnets.

TASK 2 — Topology Evidence

Star-based topology connecting the three departmental networks:



3. Network Cabling Plan

Connection	Recommended Cable	Reason	Typical Use
PC ↔ Switch	Copper straight-through, Cat6	Reliable Ethernet connection and good bandwidth	End devices
Printer/Server ↔ Switch	Copper straight-through, Cat6	Standard Ethernet connection	Network resources
Switch ↔ Router	Copper straight-through, Cat6	Ethernet uplink to routing device	Inter-network connectivity

Cat6 copper Ethernet cabling is recommended for the proposed small network because it supports common Gigabit Ethernet deployments, is practical for short indoor runs, and

is easy to terminate and troubleshoot. Fiber can be considered later if the institute expands to longer distances or requires higher-capacity uplinks.

Cabling Implementation

In Packet Tracer, we make the physical connections and record the cable type used for every link, clear from topology diagrams.

4. Switch Design and Configuration

Three access switches are used. Each switch serves one logical department. The following port allocation is a practical example and can be changed according to the number of end devices used.

Switch	Department	Example Access Ports	Uplink
SW-ADMIN	Administration	Fa0/1–Fa0/8	Fa0/24 → Router
SW-ACADEMIC	Faculty / Academic	Fa0/1–Fa0/12	Fa0/24 → Router
SW-LAB	IT / Lab	Fa0/1–Fa0/20	Fa0/24 → Router

Basic Switch Configuration Template

Use the following commands as a starting point. Device names and interface numbers should match your Packet Tracer topology.

```
enable configure
terminal hostname
SW-ADMIN no ip
domain-lookup
enable secret class
line console 0
password cisco login
exit interface range
fa0/1-8 switchport
mode access no
shutdown exit
interface fa0/24 no
shutdown exit end
copy running-config startup-config
```

Repeat the configuration for SW-ACADEMIC and SW-LAB, changing the hostname and access-port range as needed. If your instructor requires VLAN configuration, VLANs can be added as an extension; the base task mainly requires logical switch design and basic configuration.

TASK 4 — Switch Configuration and Verification

SW-ADMIN

```
SW-ADMIN#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	1	a-full	a-100	10/100BaseTX
Fa0/2		connected	1	a-full	a-100	10/100BaseTX
Fa0/3		connected	1	a-full	a-100	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX

```
SW-ADMIN#
```

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SW-ACADEMIC

SW-ACADEMIC#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	1	a-full	a-100	10/100BaseTX
Fa0/2		connected	1	a-full	a-100	10/100BaseTX
Fa0/3		connected	1	a-full	a-100	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX

SW-ACADEMIC#

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SW-LAB

SW-LAB#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	1	a-full	a-100	10/100BaseTX
Fa0/2		connected	1	a-full	a-100	10/100BaseTX
Fa0/3		connected	1	a-full	a-100	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX

SW-LAB#

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5. Router Configuration and IP Addressing

A separate IPv4 subnet is assigned to each department. This provides logical separation while allowing the router to provide communication between the segments. The /24 networks are simple to understand and leave substantial room for future devices.

Device/Interface	IP Address	Mask	Purpose
R1 G0/0	192.168.10.1	255.255.255.0	Administration gateway
R1 G0/1	192.168.20.1	255.255.255.0	Academic gateway
R1 G0/2	192.168.30.1	255.255.255.0	Lab gateway

Device	IP	Mask	Gateway
ADMIN-PC1	192.168.10.10	255.255.255.0	192.168.10.1
ADMIN-PC2	192.168.10.11	255.255.255.0	192.168.10.1
ADMIN-PC3	192.168.10.12	255.255.255.0	192.168.10.1
ACADEMIC-PC1	192.168.20.10	255.255.255.0	192.168.20.1
ACADEMIC-PC2	192.168.20.11	255.255.255.0	192.168.20.1
ACADEMIC-PC3	192.168.20.12	255.255.255.0	192.168.20.1
LAB-PC1	192.168.30.10	255.255.255.0	192.168.30.1
LAB-PC2	192.168.30.11	255.255.255.0	192.168.30.1
LAB-PC3	192.168.30.12	255.255.255.0	192.168.30.1

Segment	Network	Gateway	Usable Host Range	Broadcast
Administration	192.168.10.0/24	192.168.10.1	192.168.10.2– 192.168.10.254	192.168.10.255
Academic	192.168.20.0/24	192.168.20.1	192.168.20.2– 192.168.20.254	192.168.20.255

IT / Lab	192.168.30.0/24	192.168.30.1	192.168.30.2– 192.168.30.254	192.168.30.255
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Router-on-a-Stick Example Configuration

If one physical router interface is connected to a switch trunk, router subinterfaces can represent the three departmental networks. This example assumes the router interface connected to the switching infrastructure is GigabitEthernet0/0.

```
enable configure
terminal
hostname R1
interface gigabitEthernet0/0 no
shutdown exit interface
gigabitEthernet0/0.10
encapsulation dot1Q 10 ip address
192.168.10.1 255.255.255.0 exit
interface gigabitEthernet0/0.20
encapsulation dot1Q 20 ip address
192.168.20.1 255.255.255.0 exit
interface gigabitEthernet0/0.30
encapsulation dot1Q 30 ip address
192.168.30.1 255.255.255.0 exit
end
copy running-config startup-config
```

Because this configuration uses 802.1Q subinterfaces, the switch uplink connected to the router must be configured appropriately as a trunk, and the relevant VLANs must exist. If your lab topology instead uses separate physical router interfaces, configure each interface directly with the corresponding gateway address.

STUDENT TASK 5 — Router Configuration and Connectivity

Using IP Addressing table above:

R1 interface configuration and status:

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.10.1	YES	manual	up	up
GigabitEthernet0/1	192.168.20.1	YES	manual	up	up
GigabitEthernet0/2	192.168.30.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```
R1#
```

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Connectivity Testing between departmental networks

ADMIN-PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time=12ms TTL=127
Reply from 192.168.30.10: bytes=32 time=15ms TTL=127
Reply from 192.168.30.10: bytes=32 time=13ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 15ms, Average = 13ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=13ms TTL=127
Reply from 192.168.20.10: bytes=32 time=16ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 10ms

C:\>ping 192.168.10.1

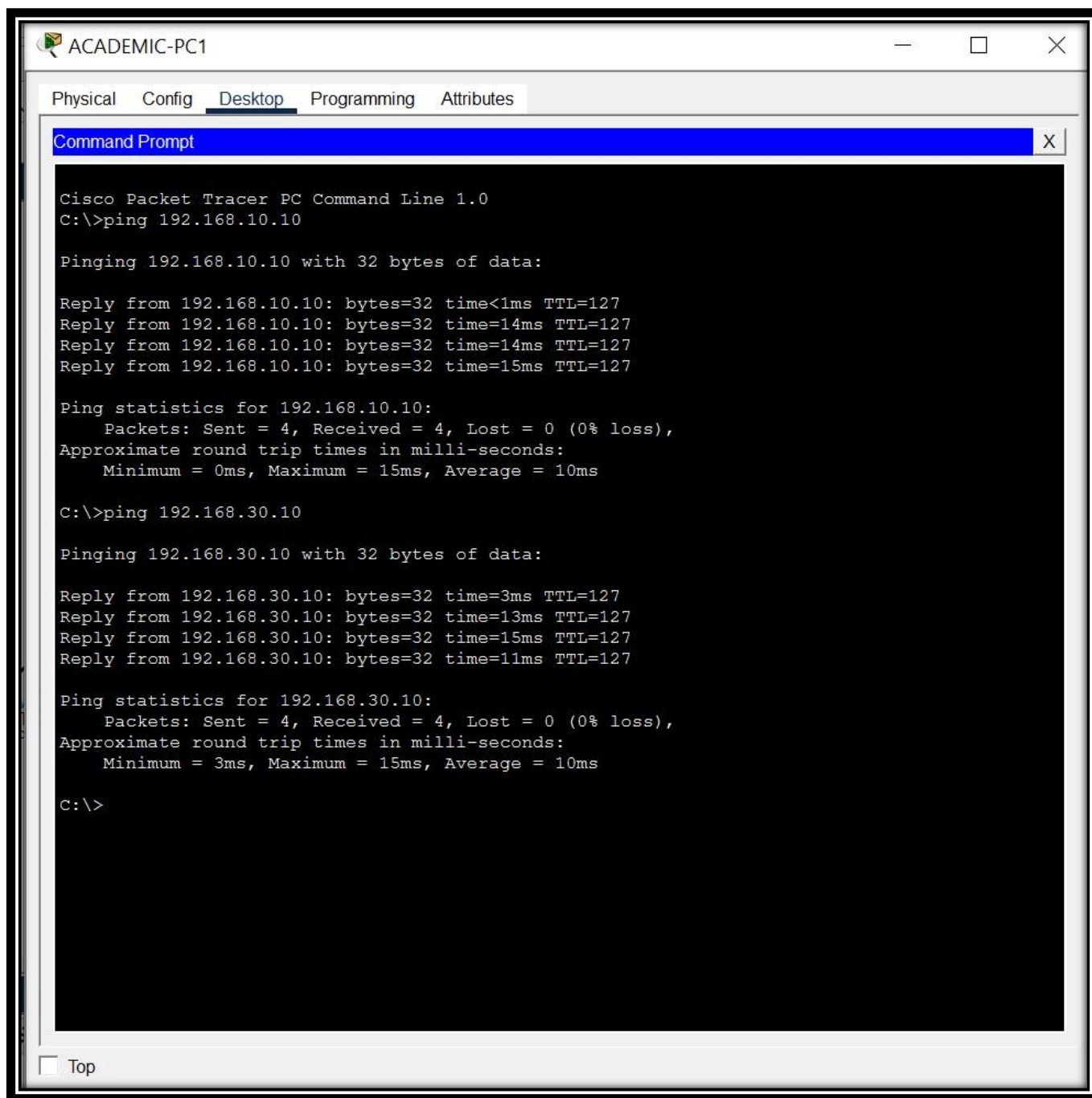
Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=3ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=10ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

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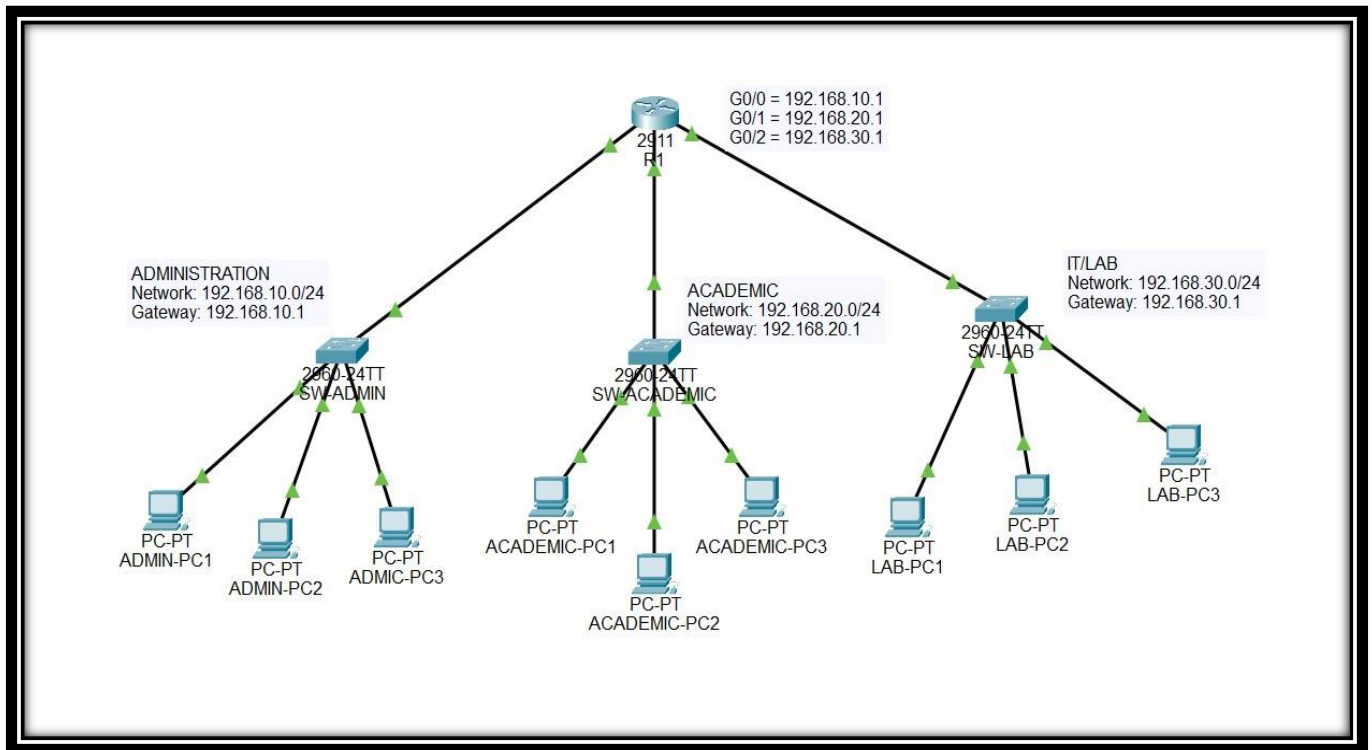


Successful inter-network connectivity testing using ping

6. Complete Logical Network Diagram

TASK 6 — Complete Logical Network Diagram

Final Network Diagram



7. Future Improvements

- VLAN-based segmentation and inter-VLAN security policies can be expanded to isolate departments and reduce unnecessary broadcast traffic.
- Wireless access points can be added to provide managed Wi-Fi for students, faculty, and staff, with separate SSIDs and security policies.
- A firewall or dedicated security appliance can be introduced between the internal network and the Internet to improve perimeter security.
- Redundant switches, links, or routing equipment can be considered to improve availability as the institute grows.
- Centralized services such as DHCP, DNS, directory services, monitoring, and backup can be deployed for easier administration.
- Fiber-optic uplinks can be introduced for longer distances or higher-capacity backbone connections.

TASK 7 — Future Improvements

Add at least three future improvements in your own words. Explain why each improvement would be useful for the institute and how it would affect scalability, security, reliability, or performance.

1. Wireless access points can be added in the future to provide W-Fi connectivity for students, teachers, and administrative staff.
2. A firewall can be added between the router and the internet to improve network security and control incoming and outgoing traffic.
3. Additional switches, PCs, and higher-capacity links can be added as the institute grows. Fiber-optic connections can also be considered for longer-distance or higher-speed communications.

8. Testing and Verification

After implementation, verify that all intended links are operational and that hosts can communicate with their default gateway. Then verify communication between at least two different departmental networks. Useful Cisco IOS commands include:

```
show ip interface
brief show running-
config show vlan
brief show interfaces
trunk ping
<destination-IP>
```

TASK 8 — Test Results

Source Device	Destination	Test / Command	Result
ADMIN-PC1	192.168.10.1	Ping	Successful
ADMIN-PC1	192.168.20.10	Ping	Successful
ADMIN-PC1	192.168.30.10	Ping	Successful
ACADEMIC-PC1	192.168.10.10	Ping	Successful
ACADEMIC-PC1	192.168.30.10	Ping	Successful
LAB-PC1	192.168.10.10	Ping	Successful
LAB-PC1	192.168.20.10	Ping	Successful

9. Discussion

The Packet Tracer implementation successfully demonstrated communication between the Administration, Academic, and IT/Lab networks using the configured router interfaces and IPv4 addressing scheme.

The star-based physical arrangement also allows an individual endpoint or access link to be serviced without redesigning the whole network. The practical configuration must, however, be verified in the chosen simulator or hardware environment, especially interface names, VLAN/trunk requirements, and successful end-to-end connectivity.

10. Conclusion

This open-ended laboratory exercise develops a complete small-network design for an educational institute. The report covers requirements estimation, topology selection, cabling, switch planning, IP addressing, router configuration, network diagram requirements, testing, and future expansion. Packet Tracer implementation and observations ensure that the final submission contains authentic configuration evidence and verified connectivity results. Thank you!